Announced Rescue Pricing with Latent Market Thickness

Group-meeting note — benchmark, numerical pattern, and formal model

27 August 2026

Preview. A publicly known single driver gives an exact cutoff-WPBE and a closed-form optimal menu. With latent Poisson supply, the completion value of rescue pricing is small in thin markets, largest at an intermediate thickness, and small again in thick markets. For $(\alpha, \beta, \delta, \gamma) = (1, .8, .8, 0)$, the deterministic numerical solution peaks near $m = 9.72$ and improves optimized completion by 6.16 percentage points. The endpoint pattern is proved; strict single-peakedness is a conjecture.

1 Exact one-rider/one-driver benchmark

1.1 Closed-form solution

The rider has $v \sim U[0,1]$, the driver has $c \sim U[0,1]$, and the platform maximizes completion. Delayed rider value is βv and delayed incumbent surplus is $\delta(p_j - c)$. The platform commits to $0 \le p_1 \le p_2 \le 1$. The number of drivers is publicly known to be exactly one; there is no survival shock or fresh entry.

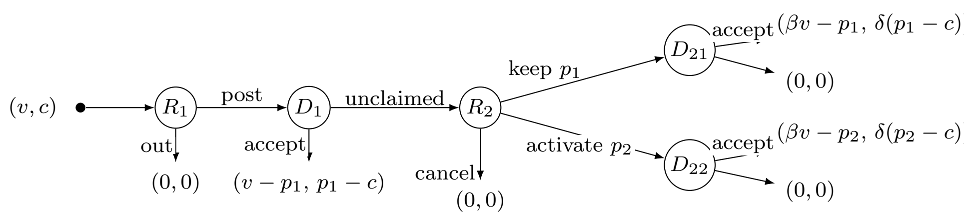

Figure 1: Extensive form.

For an active menu $0 \le p_1 < p_2 < \beta$, the driver accepts immediately iff $c \le \widehat{c}$. After rejection, the rider cancels, repeats p_1, or activates p_2 according to

$$v < \frac{p_1}{\beta}, \qquad \frac{p_1}{\beta} \le v \le v^M(\widehat{c}), \qquad v > v^M(\widehat{c}), \qquad v^M(\widehat{c}) = \frac{p_1 + p_2 - \widehat{c}}{\beta}.$$

The marginal driver's accept-versus-wait indifference gives the WPBE cutoff

$$\widehat{c}(p_1, p_2) = \left[p_1 - \frac{\delta(p_2 - p_1)(\beta - p_2)}{\beta(1 - p_1) - \delta(\beta - p_2)} \right]_+, \tag{1}$$

and equilibrium completion is

$$M_1(p_1, p_2) = (1 - p_1)\widehat{c} + \frac{(\beta - p_2)(p_2 - \widehat{c})}{\beta}. \tag{2}$$

Thus the cutoff is calculated from the WPBE; the platform then optimizes over menus subject to this equilibrium response.

Fixing a target cutoff reduces the two-price problem to one dimension. The unique continuation envelope and its implementing first price are

$$\widehat{p}_2(\widehat{c}) = \frac{\beta + \widehat{c}}{2}, \qquad \widehat{p}_1(\widehat{c}) = \frac{1 + \widehat{c} - \sqrt{\Delta(\widehat{c})}}{2}, \qquad \Delta(\widehat{c}) = (1 - \widehat{c})^2 - \frac{\delta}{\beta}(\beta - \widehat{c})^2. \tag{3}$$

Substitution into (2) gives

$$M_1^* = \max\left\{ \frac{1}{4},\ \max_{\widehat{c} \in [0,\beta]} J_\delta(\widehat{c}) \right\}, \quad J_\delta(\widehat{c}) = \frac{\widehat{c}[1 - \widehat{c} + \sqrt{\Delta(\widehat{c})}]}{2} + \frac{(\beta - \widehat{c})^2}{4\beta}. \tag{4}$$

If $\beta \le 1/2$ or $\delta = 1$, the optimal value is the flat benchmark $1/4$. If $\beta > 1/2$ and $\delta < 1$, J_δ is strictly concave and has a unique interior maximizer, which implements $p_1^* < p_2^*$.

1.2 Numerical illustration

At $\beta = \delta = 4/5$, the first-order condition becomes $12 - 20\widehat{c} = 5\widehat{c}\sqrt{9 - 10\widehat{c}}$. Squaring yields $(5\widehat{c} - 4)(50\widehat{c}^2 + 75\widehat{c} - 36) = 0$; checking the unsquared equation and feasibility leaves the positive root of the quadratic. Its discriminant is $75^2 + 4(50)(36) = 225 \cdot 57$, which explains the appearance of $\sqrt{57}$:

$$\widehat{c}^* = \frac{3(\sqrt{57} - 5)}{20} = .382475, \quad p_1^* = \frac{11 + \sqrt{57}}{40} = .463746, \quad p_2^* = \frac{1 + 3\sqrt{57}}{40} = .591238.$$

The optimal rescue menu completes 25.9581% of requests, compared with 25.0000% under the best flat price.

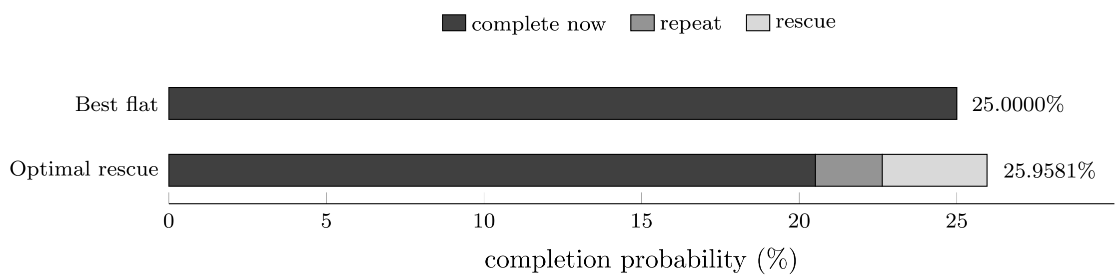

Figure 2: Exact-one-driver completion at $\beta = \delta = .8$. The 0.9581 pp gain combines less completion now with repeat and rescue completion later.

β	δ	$\widehat{c}^*$	p_1^*	p_2^*	M_1^*
.6	.5	.460069	.467733	.530034	.253038
.6	.8	.478613	.488210	.539306	.251089
.8	.5	.370421	.420148	.585210	.272458
.8	.8	.382475	.463746	.591238	.259581
.9	.5	.329857	.405843	.614928	.286282
.9	.8	.314018	.453693	.607009	.266932

Each row re-solves the WPBE and optimal menu. More patient riders support a larger rescue region; less patient incumbents make waiting less attractive and increase the completion value of escalation.

2 From one driver to latent market thickness

2.1 Thickness mechanism and prior

Now only the public mean m is known: $N^I \sim \mathrm{Pois}(m)$ is latent. Therefore $m = 1$ is not the publicly known $N = 1$ benchmark. At the same m, define

$$M_R^*(m) = \max_{0 \le p_1 \le p_2 \le 1} M_m(p_1, p_2), \qquad M_F^*(m) = \max_{p \in [0,1]} M_m(p, p), \qquad V(m) = M_R^*(m) - M_F^*(m).$$

Here M_m is equilibrium completion. Thus $100V(m)$ is the percentage-point gain from allowing the best rescue menu instead of the best flat price in the *same* market. It is not the marginal value of adding drivers.

Region	Initial supply	Failure and backup pool	Prior for $V(m)$
Thin m	Usually no incumbent	Little supply exists to rescue	Small; first order in m
Middle m	Initial failure remains material	Failure screens incumbents, while a useful backup pool often remains	Potentially largest
Thick m	Many potential acceptors	Optimized flat pricing already completes most requests	Small; tends to zero

The middle region contains the trade-off: a larger pool raises terminal coverage, but the announced p_2 also induces more incumbents to wait. The direction is therefore an equilibrium-design question rather than a mechanical coverage comparison.

2.2 Deterministic numerical evidence

The reported simulation is deterministic equilibrium optimization, not Monte Carlo. For every m, the procedure solves the unique cutoff-WPBE for each candidate menu, reduces the active branch to one scalar cutoff, checks all stationary points and boundaries, and compares the result with the optimized flat price. The baseline is

$$F = G = U[0,1], \qquad (\alpha, \beta, \delta, \gamma) = (1, .8, .8, 0).$$

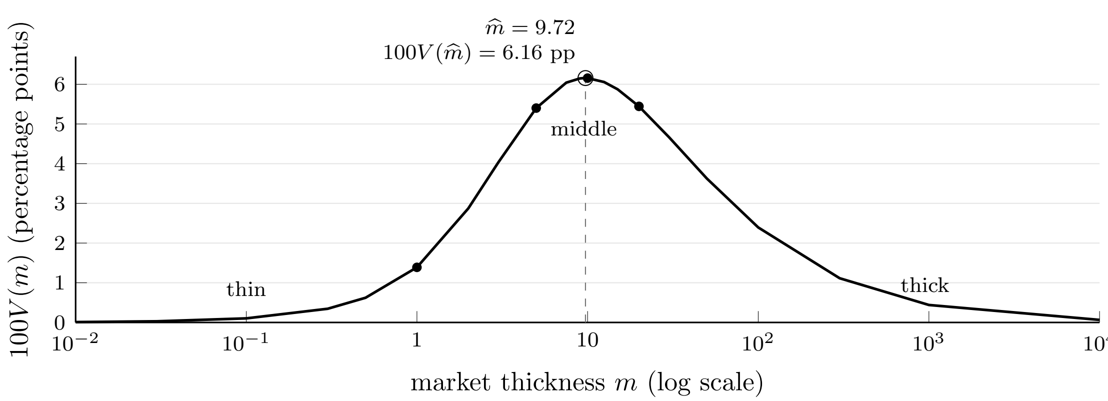

Figure 3: Optimized rescue value across thickness. Positivity, continuity, and both endpoint limits are analytical; the connected values and apparent single hump are numerical.

m	$\widehat{c}^*$	p_1^*	p_2^*	M_R^*	M_F^*	$100V(m)$
1	.320340	.391928	.546378	.213230	.199346	1.3883 pp
5	.197789	.236946	.413098	.597813	.543810	5.4003 pp
10	.132863	.155922	.310306	.755154	.693602	6.1552 pp
20	.080430	.092538	.208054	.862392	.807956	5.4437 pp

The largest value located is at $(\widehat{m}, \widehat{c}^*, p_1^*, p_2^*) = (9.718617, .135370, .159002, .314745)$. Across the four highlighted markets, the selected cutoff and both prices fall, yet rescue coverage rises because the platform draws from a larger pool.

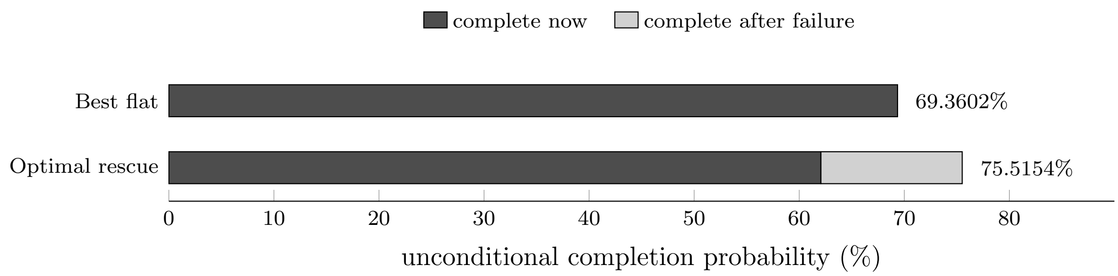

Figure 4: At $m = 10$, rescue sacrifices 7.3070 pp of immediate completion and recovers 13.4622 pp after failure, for a net gain of 6.1552 pp.

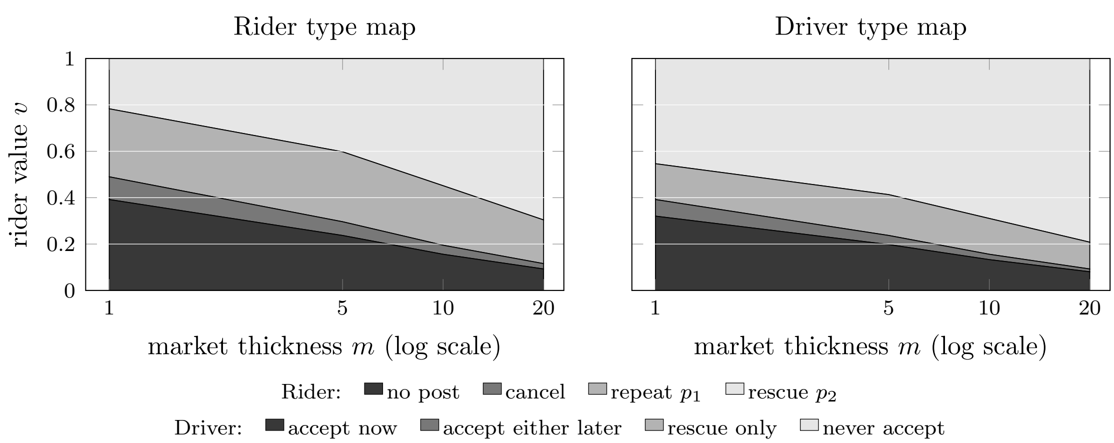

Figure 5: Type regions at $m = 1, 5, 10, 20$. Each vertical slice partitions $[0, 1]$. Rider rescue mass grows from .2169 to .6960, while pool-level rescue coverage rises from .2023 to .9221. Type shares are not path probabilities.

3 Formal thickness model

3.1 Primitives and timing

The numerical pattern motivates a model that separates the size and composition of continuation supply.

Param.	Domain	Economic object	Baseline numerics
m	$(0, \infty)$	Public mean incumbent supply, $N^I \sim \text{Pois}(m)$	varied
α	$[0, 1]$	Survival probability of a waiting incumbent	1
γ	$[0, \infty)$	Fresh-entry intensity, $N^E \sim \text{Pois}(\gamma m)$	0
β	$(0, 1]$	Rider discount factor	.8
δ	$(0, 1]$	Incumbent discount factor	.8
F	continuous CDF	Persistent incumbent and entrant cost distribution	$U[0, 1]$
G	continuous CDF	Rider value distribution	$U[0, 1]$

All primitives and the committed menu are common knowledge; realized counts and types are latent except to the driver who owns each type.

Stage 1. Commitment. The platform observes m and commits to $0 \leq p_1 \leq p_2 \leq 1$; p_2 is available only after universal rejection.

Stage 2. Rider entry. The rider observes $v \sim G$ and chooses whether to post.

Stage 3. Incumbent response. A latent $N^I \sim \text{Pois}(m)$ pool draws persistent costs $c_i \sim F$. Drivers simultaneously accept p_1 or wait without observing pool size or rival actions.

Stage 4. Assignment or public failure. One acceptor is selected uniformly. Otherwise, universal rejection becomes public but does not reveal the count or costs.

Stage 5. Rider continuation. The rider cancels, repeats p_1, or activates p_2 using v, the menu, and the failure posterior.

Stage 6. Survival, entry, and terminal assignment. Each waiting incumbent survives independently with probability α; independently, $N^E \sim \text{Pois}(\gamma m)$ fresh drivers arrive. At the active p_j, one acceptor is selected uniformly. A selected incumbent receives $\delta(p_j - c_i)$ and a selected entrant receives $p_j - c_i$.

The baseline figures set $\gamma = 0$ to isolate strategic waiting. The formal extension retains both screened surviving incumbents and unscreened fresh entrants; they cannot generally be collapsed into a single thickness scalar.

3.2 Cutoff-WPBE and menu solution

In an anonymous symmetric pure-cutoff WPBE, incumbents accept p_1 iff $c_i \leq \widehat{c}$. A candidate cutoff therefore determines both the failure posterior and the supply available after failure. Poisson thinning gives

$$\phi(x) = \frac{1 - e^{-x}}{x}, \quad \phi(0) = 1, \qquad \lambda_j(\widehat{c}) = m\{\alpha[F(p_j) - F(\widehat{c})] + \gamma F(p_j)\}, \qquad C_j(\widehat{c}) = 1 - e^{-\lambda_j(\widehat{c})}.$$

Here ϕ is a focal driver's expected assignment share. In λ_j, $m\alpha[F(p_j) - F(\widehat{c})]$ is the screened mass of surviving incumbents and $m\gamma F(p_j)$ is the unscreened entrant mass. Their sum determines terminal coverage C_j.

Rider continuation. The rider posts iff $v \geq p_1$. After universal rejection, she compares

$$0, \qquad C_1(\widehat{c})(\beta v - p_1), \qquad C_2(\widehat{c})(\beta v - p_2).$$

When $C_2(\widehat{c}) > C_1(\widehat{c})$, repeat and rescue are equally valuable at

$$v^M(\widehat{c}) = \frac{p_2 C_2(\widehat{c}) - p_1 C_1(\widehat{c})}{\beta[C_2(\widehat{c}) - C_1(\widehat{c})]}. \tag{5}$$

Thus the conditional rider masses are

$$\eta^{\mathrm{rep}}(\widehat{c}) = \Pr\left\{ \frac{p_1}{\beta} \leq v \leq v^M(\widehat{c}) \,\middle|\, v \geq p_1 \right\}, \qquad \eta^{\mathrm{res}}(\widehat{c}) = \Pr\left\{ v > v^M(\widehat{c}) \,\middle|\, v \geq p_1 \right\},$$

with empty regions assigned zero mass. If $C_2 = C_1$, rescue adds no coverage and is unused.

Driver cutoff. Put $\eta^1 = \eta^{\mathrm{rep}}$ and $\eta^2 = \eta^{\mathrm{res}}$. A focal incumbent of type c compares immediate acceptance with discounted waiting:

$$\begin{aligned} A(c; \widehat{c}) &= \phi(mF(\widehat{c}))(p_1 - c), \\ W(c; \widehat{c}) &= \delta\alpha e^{-mF(\widehat{c})} \sum_{j=1}^{2} \eta^j(\widehat{c}) \phi(\lambda_j(\widehat{c}))(p_j - c)^+. \end{aligned} \tag{6}$$

The factors in W are, in order, incumbent discounting, own survival, universal rejection by rivals, the probability of each rider continuation action, terminal assignment share, and terminal surplus. An interior cutoff solves $A(\widehat{c}; \widehat{c}) = W(\widehat{c}; \widehat{c})$; one-sided inequalities apply at 0 and p_1. In the uniform no-entry model this equation has a unique cutoff value for every menu.

Completion and platform objective. Once the cutoff and rider regions have been recovered, completion is

$$M_m(p_1, p_2; \widehat{c}) = (1 - G(p_1)) \left[1 - e^{-mF(\widehat{c})} + e^{-mF(\widehat{c})} \{ \eta^{\mathrm{rep}} C_1 + \eta^{\mathrm{res}} C_2 \} \right]. \tag{7}$$

The terms are posting, first-window completion, and failure followed by terminal completion. The platform maximizes (7) only after imposing (5)–(6); this is the WPBE-constrained menu-design problem.

For $F = G = U[0,1]$ and $\gamma = 0$, put $k = \alpha m$ and

$$S_m(\widehat{c}, p_2) = (\beta - p_2)\{1 - e^{-k(p_2 - \widehat{c})}\}, \qquad h_m(\widehat{c}) = \frac{e^{m\widehat{c}} - 1}{\widehat{c}}, \quad h_m(0) = m.$$

The active branch admits an exact three-step reduction:

1. **Given $\widehat{c}$, solve p_2.** Rider switching makes continuation completion telescope to $S_m(\widehat{c}, p_2)/\beta$. Its unique maximizer solves

$$e^{k(\widehat{p}_2 - \widehat{c})} - 1 = k(\beta - \widehat{p}_2), \qquad \widehat{p}_2 = \beta - \frac{W_0(e^{1 + k(\beta - \widehat{c})}) - 1}{k}.$$

2. **Use driver indifference to solve p_1.** With $p_2 = \widehat{p}_2$, equation (6) becomes

$$(1 - p_1)(p_1 - \widehat{c}) = \frac{\delta S_m(\widehat{c}, \widehat{p}_2)}{\beta h_m(\widehat{c})}.$$

The root satisfying $\widehat{c} \leq p_1 \leq \widehat{p}_2$ is

$$\widehat{p}_1^{\delta} = \frac{1 + \widehat{c} - \sqrt{(1 - \widehat{c})^2 - \dfrac{4\delta S_m(\widehat{c}, \widehat{p}_2)}{\beta h_m(\widehat{c})}}}{2}.$$

3. Optimize the remaining cutoff. Substitution into (7) leaves

$$J_{\delta,m}(\widehat{c}) = [1-\widehat{p}_1^{\delta}](1-e^{-m\widehat{c}}) + e^{-m\widehat{c}}\frac{S_m(\widehat{c},\widehat{p}_2)}{\beta}, \qquad \widehat{c}\in[0,\beta].$$

Consequently,

$$\widehat{c}^*(m) \in \arg\max_{\widehat{c}\in[0,\beta]} J_{\delta,m}(\widehat{c}), \qquad p_1^*(m) = \widehat{p}_1^{\delta}(\widehat{c}^*(m);m), \qquad p_2^*(m) = \widehat{p}_2(\widehat{c}^*(m);m).$$

Thus both prices are obtained from the WPBE, not from an unconstrained price grid. The optimized flat benchmark has

$$p_F^*(m) = \frac{1+m-W_0(e^{1+m})}{m}.$$

4 Evidence boundary, extensions, and literature

4.1 Established results and conjectures

Status	Statement
Established	In the uniform no-entry model, every menu has a unique cutoff value in the maintained WPBE class and the optimal menu reduces to the scalar problem above. For $\beta \geq 1/2$, $V(m) > 0$ at every finite m, V is continuous, and $V(m) \to 0$ as $m \downarrow 0$ and $m \to \infty$. Hence at least one positive finite maximizer exists.
Established limits	With $\alpha = 1$, $\gamma = 0$, and $\beta = \delta = .8$, $$\lim_{m\downarrow 0}\frac{V(m)}{m} = M_1^* - \frac{1}{4} = \frac{461-57\sqrt{57}}{3200} = .00958107.$$ Thus the exact-one-driver gain is the thin-market coefficient. For fixed positive α,β and $\gamma = 0$, $V(m) \sim (\log m)/m$ in thick markets.
Numerical evidence	The baseline curve has one located peak $\widehat{m} = 9.718617$ with a 6.1566 pp gain. At $m = 1, 5, 10, 20$, the selected $\widehat{c}^*, p_1^*, p_2^*$ fall, while rider rescue mass and pool-level rescue coverage rise.
Conjectures	For $\beta \geq 1/2$ in the uniform no-entry model, $V(m)$ may be strictly single-peaked with a unique maximizer. A monotone selection of $\widehat{c}^*(m), p_1^*(m), p_2^*(m)$ may also exist, but neither optimizer uniqueness nor global monotonicity is proved. For $\beta < 1/2$, an initial zero plateau already rules out strict single-peakedness in this form.
Mixed supply	With $\alpha < 1$ and $\gamma > 0$, a small rescue increment strictly improves an optimized flat policy under an explicit rider-activity condition; when the condition fails strictly, the increment is unused. Holding continuation headcount or price-marginal capacity fixed, a larger incumbent share locally intensifies cutoff erosion. Global menu design and the full shape of $V_{\alpha,\gamma}(m)$ remain open; composition effects on completion can be nonmonotone.

4.2 Position relative to the closest literature

Literature	Existing mechanism	Additional object here
Li and Kuo (2013)	Dutch clock with uncertain bidder count	Private rider continuation and completion design
Chen (2012); Correa et al. (2016)	Strategic waiting and contingent preannouncement	Drivers wait for higher pay under assignment competition
Hu et al. (2022)	Two-sided temporal response to surge pricing	Same-request failure-contingent commitment and posterior updating
Lee and Li (2023)	Screened incumbents and later entrants in search	Posted payments with abandon, repeat, and rescue
Loertscher et al. (2022)	Optimal accumulation of market thickness	Value of rescue pricing while holding the same m fixed

The contribution is not two prices, Poisson supply, or strategic waiting separately. It is their joint cutoff-WPBE design with latent realized competition, universal-rejection learning, private rider continuation, screened incumbents, unscreened entrants, and a completion objective.

Empirical calibration placeholder. A request-level calibration needs the announced (p_1, p_2), a public pre-window thickness proxy, initial exposures and universal rejection, rider continuation, driver identities linking survivors to the first window, fresh entrant exposure, and final completion. These fields identify or discipline $(m, \alpha, \gamma, F, G)$ and the dynamic responses governed by (β, δ). The paper specifies this mapping but does not insert synthetic estimates as empirical findings.

References.

Chen, C.-H. 2012. Name your own price at Priceline.com: Strategic bidding and lockout periods. *Review of Economic Studies* 79(4), 1341–1369.

Correa, J., R. Montoya, C. Thraves. 2016. Contingent preannounced pricing policies with strategic consumers. *Operations Research* 64(1), 251–272.

Hu, B., M. Hu, H. Zhu. 2022. Surge pricing and two-sided temporal responses in ride hailing. *M&SOM* 24(1), 91–109.

Lee, J., D. Li. 2023. Seller compound search for bidders. *Journal of Industrial Economics* 71(4), 1004–1037.

Li, Z., C.-C. Kuo. 2013. Design of discrete Dutch auctions with an uncertain number of bidders. *Annals of Operations Research* 211, 255–272.

Loertscher, S., E. Muir, P. G. Taylor. 2022. Optimal market thickness. *Journal of Economic Theory* 200, 105383.